

EXERCISE 1

$n = 20$ students

simple Gaussian LM: $Y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$

- a) assumption on ε_i : $\varepsilon_i \sim N(0, \sigma^2)$ iid for $i = 1, \dots, 20$
(normality, mean = 0, homoscedasticity, independence)

- b) The MLE of (β_1, β_2) are

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}$$

$$\hat{\beta}_2 = \frac{\text{cov}(x, y)}{\text{var}(x)} = \frac{s_{xy}}{s_x^2}$$

First, we compute $\hat{\beta}_2$:

$$\begin{aligned} n \cdot \text{cov}(x, y) &= \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \\ &= \sum_{i=1}^n (x_i y_i - \bar{x} y_i - x_i \bar{y} + \bar{x} \bar{y}) = \sum_{i=1}^n x_i y_i - n \bar{x} \bar{y} \end{aligned}$$

$$\Rightarrow n \cdot \text{cov}(x, y) = 257.66 - 20 \cdot \frac{100}{20} \cdot \frac{50}{20} = 7.66$$

$$n \cdot \text{var}(x) = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n \bar{x}^2$$

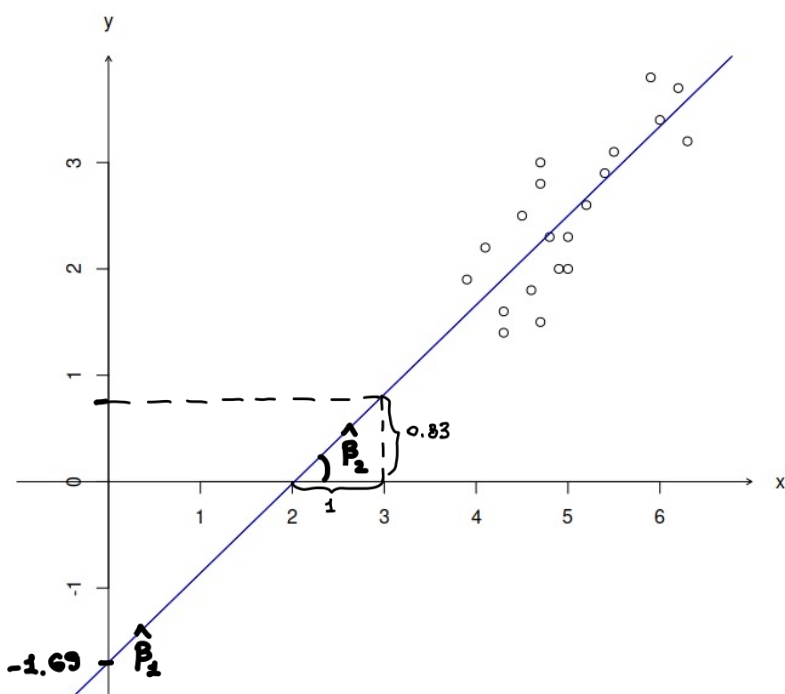
$$\Rightarrow n \text{ var}(x) = 509.12 - 20 \cdot \left(\frac{100}{20}\right)^2 = 9.12$$

$$\text{Hence } \hat{\beta}_2 = \frac{7.66}{9.12} = 0.8399$$

$$\text{Then, } \hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x} = \frac{50}{20} - 0.8399 \cdot \frac{100}{20} = -1.6995$$

- $\hat{\beta}_1$ is the mean GPA corresponding to an entrance test score equal to 0
- $\hat{\beta}_2$ is the expected change in the mean GPA if I increase the entrance test score of 1 unit.

c) $\hat{y}_i = -1.6995 + 0.8399 x_i$



- d) $\hat{B}_2(Y)$: is the random interval defined by
 $P(\beta_2 \in \hat{B}_2(Y)) = 0.99 \quad \forall \beta_2 \in \mathbb{R}$

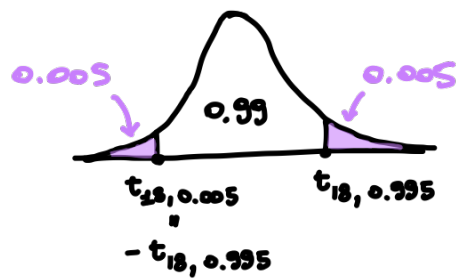
To obtain it I use the pivotal quantity

$$T = \frac{\hat{\beta}_2(Y) - \beta_2}{\sqrt{\hat{\text{var}}(\hat{\beta}_2)}} \sim t_{19}$$

$$P(-t_{19, 0.995} < T < t_{19, 0.995}) = 0.99$$

$$P(-t_{19, 0.995} < \frac{\hat{\beta}_2(Y) - \beta_2}{\sqrt{\hat{\text{var}}(\hat{\beta}_2)}} < t_{19, 0.995}) = 0.99$$

$$\text{Hence } \beta_2 \in \hat{\beta}_2 \pm t_{19, 0.995} \cdot \sqrt{\hat{\text{var}}(\hat{\beta}_2)}$$



$$\beta_2 \in (0.8399 \pm 2.878 \cdot 0.1440) = (0.4254; 1.2543) \quad \text{it does not contain zero.}$$

If the CI contains 0, it means that we do not reject the hypothesis

$H_0: \beta_2 = 0$ vs $H_1: \beta_2 \neq 0$ using a significance level 0.01.

Hence, in this case, it would suggest that the entrance score is not a good measure for predicting the students' GPA

- e) Yes, because $0 \notin \text{CI}$ in the previous point.

- f) Prediction intervals get wider when we try to predict y far from the observed range of points x_i . In particular, the smallest interval is obtained at $x_i = \bar{x}$.

Since $\bar{x} = 5.0 = x_A$ (entrance test of student A)

while $x_B = 6.5$ is even outside of the observed range,

the prediction interval for student B will be wider.

EXERCISE 2

Y_i = survival of individual i $i = 1, \dots, 714$

a) the model is

$$Y_i \sim \text{Bernoulli}(\pi_i) \quad \text{independent} \quad i = 1, \dots, 714$$

$$\text{logit}(\pi_i) = \beta_1 + \beta_2 \underbrace{\mathbb{1}(\text{class}_i = \text{first})}_{\text{dummy} = \begin{cases} 1 & \text{if class = first} \\ 0 & \text{if class} \neq \text{first} \end{cases}} + \beta_3 \underbrace{\mathbb{1}(\text{gender}_i = \text{man})}_{\text{dummy} = \begin{cases} 1 & \text{if gender}_i = \text{man} \\ 0 & \text{if gender}_i = \text{woman} \end{cases}} + \beta_4 \text{age}_i$$

The estimated model is

$$Y_i \sim \text{Bernoulli}(\hat{\pi}_i) \quad \text{logit}(\hat{\pi}_i) = \text{log} \frac{\hat{\pi}_i}{1 - \hat{\pi}_i} = 1.50 + 2.01 \mathbb{1}(\text{class}_i = \text{first}) - 2.54 \mathbb{1}(\text{gender}_i = \text{man}) - 0.029 \cdot \text{age}_i$$

$$b) \quad P(Y_i; \pi_i) = \pi_i^{y_i} (1 - \pi_i)^{1 - y_i}$$

$$P(y_1, \dots, y_n; \pi) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1 - y_i}$$

likelihood function

$$L(\pi) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1 - y_i}$$

$$L(\beta) = \prod_{i=1}^n \left\{ \left(\frac{e^{\tilde{x}_i^T \beta}}{1 + e^{\tilde{x}_i^T \beta}} \right)^{y_i} \left(\frac{1}{1 + e^{\tilde{x}_i^T \beta}} \right)^{1 - y_i} \right\} \quad \text{where I denote with } \tilde{x}_i^T \text{ the row vector of the observed values of the covariates for individual } i$$

the loglikelihood is

$$\begin{aligned} \ell(\beta) &= \log L(\beta) \\ &= \sum_{i=1}^n \left\{ \tilde{x}_i^T \beta y_i - y_i \log(1 + e^{\tilde{x}_i^T \beta}) - (1 - y_i) \log(1 + e^{\tilde{x}_i^T \beta}) \right\} \\ &= \sum_{i=1}^n \left\{ \tilde{x}_i^T \beta y_i - y_i \log(1 + e^{\tilde{x}_i^T \beta}) - \log(1 + e^{\tilde{x}_i^T \beta}) + y_i \log(1 + e^{\tilde{x}_i^T \beta}) \right\} \\ &= \sum_{i=1}^n \left\{ \tilde{x}_i^T \beta y_i - \log(1 + e^{\tilde{x}_i^T \beta}) \right\} \end{aligned}$$

c) 1. z value of "Gender"

this is the observed value of the test statistic for testing

$$H_0: \beta_3 = 0 \quad \text{vs} \quad H_1: \beta_3 \neq 0$$

we use the statistic

$$z = \frac{\hat{\beta}_3 - 0}{\sqrt{[\hat{J}(\hat{\beta})]_{3,3}}} \stackrel{H_0}{\sim} N(0, 1)$$

$$\text{hence the needed quantity is } z^{\text{obs}} = \frac{-2.5473}{0.2017} = -12.629$$

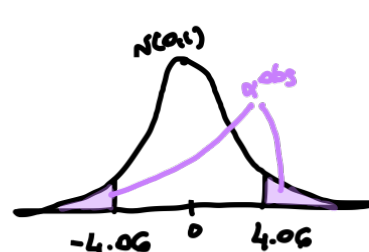
2. $P(|z| > 1.96)$ for "Age"

this is the value of the test $H_0: \beta_4 = 0$ vs $H_1: \beta_4 \neq 0$

Similar to the previous point, it is based on the quantity

$$z = \frac{\hat{\beta}_4}{\sqrt{[\hat{J}(\hat{\beta})]_{4,4}}} \stackrel{H_0}{\sim} N(0, 1)$$

$$\begin{aligned} \text{Hence the value is } \alpha^{\text{obs}} &= P_{H_0}(|z| > |z^{\text{obs}}|) \\ &= 2 P_{H_0}(z > |z^{\text{obs}}|) \\ &= 2 (1 - \Phi(|z^{\text{obs}}|)) \\ &= 2 (1 - \Phi(4.06)) \end{aligned}$$



$$\Phi(4.06) \approx 1 \Rightarrow \alpha^{\text{obs}} \approx 0$$

3. All of them.

$$d) \quad \hat{\pi}_A = (\hat{\pi}_A | \text{gender}_A = 0, \text{class}_A = 1, \text{age}_A = 30) = \frac{e^{\tilde{x}_A^T \hat{\beta}}}{1 + e^{\tilde{x}_A^T \hat{\beta}}} = 0.9317$$

$$\text{since } \tilde{x}_A^T \hat{\beta} = 1.5003 + 2.0103 \cdot 1 - 2.5473 \cdot 0 - 0.0299 \cdot 30 = 2.6136$$

$$\Rightarrow \frac{e^{\tilde{x}_A^T \hat{\beta}}}{1 + e^{\tilde{x}_A^T \hat{\beta}}} = \frac{e^{2.6136}}{1 + e^{2.6136}} = 0.9317$$

$$\text{the odds for individual A are } \frac{\hat{\pi}_A}{1 - \hat{\pi}_A} = 13.648$$

we know that

$$\text{logit } \hat{\pi}_A = \hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_4 \cdot 30$$

$$\text{logit } \hat{\pi}_B = \hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_4 \cdot 31 = \hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_4 \cdot 30 + \hat{\beta}_4$$

$$\begin{aligned} \text{Hence } \text{logit } \hat{\pi}_B - \text{logit } \hat{\pi}_A &= \hat{\beta}_4 \\ \log \frac{\hat{\pi}_B}{1 - \hat{\pi}_B} - \log \frac{\hat{\pi}_A}{1 - \hat{\pi}_A} &= \log \frac{\frac{\hat{\pi}_B}{1 - \hat{\pi}_B}}{\frac{\hat{\pi}_A}{1 - \hat{\pi}_A}} = \hat{\beta}_4 \end{aligned}$$

$$\Rightarrow \frac{\frac{\hat{\pi}_B}{1 - \hat{\pi}_B}}{\frac{\hat{\pi}_A}{1 - \hat{\pi}_A}} = e^{\hat{\beta}_4}$$

The odds of individual A change of a multiplicative factor $e^{\hat{\beta}_4}$ to obtain the odds of individual B.

$$\text{This is, } \text{odds}_B = \frac{\hat{\pi}_B}{1 - \hat{\pi}_B} = e^{-0.0299} \cdot 13.648 = 13.246$$

e) "class" is a dummy variable

Hence

$$\hat{\beta}_2 = \log \frac{\frac{P(Y_i = 1 | \text{class}_i = 1)}{P(Y_i = 0 | \text{class}_i = 1)}}{\frac{P(Y_i = 1 | \text{class}_i = 0)}{P(Y_i = 0 | \text{class}_i = 0)}} \Rightarrow e^{\hat{\beta}_2} = \frac{(\text{odds} | \text{class}_i = 1)}{(\text{odds} | \text{class}_i = 0)}$$

The odds of a person in third class are multiplied by $e^{\hat{\beta}_2} = e^{2.0103} = 7.4657$ to obtain the odds of a person in first class (keeping the other covariates fixed).

Since $\hat{\beta}_2$ is positive, a person with a first-class ticket has a higher probability of surviving, compared to a person with a second or third-class ticket.

$$f) \quad \begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 < 0 \end{cases}$$

we use the test statistic

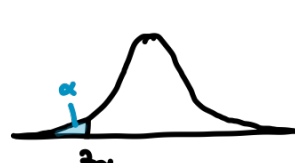
$$z = \frac{\hat{\beta}_2 - 0}{\text{se}(\hat{\beta}_2)} \stackrel{H_0}{\sim} N(0, 1)$$

the reject region is for large negative values

using a significance level α : reject if $z^{\text{obs}} < z_\alpha$

$$z^{\text{obs}} = 8.11 \quad (\text{in the table})$$

Hence I do not reject H_0 for all used α



g) the residual deviance is the log-likelihood test between the saturated model and the proposed model:

$$D(\text{model}) = 2 \{ \tilde{\ell}(\text{saturated}) - \hat{\ell}(\text{model}) \}$$

where $\tilde{\ell}(\text{saturated})$ is the maximum of the log-likelihood under a model with n parameters,

and $\hat{\ell}(\text{model})$ is the maximum of the log-likelihood under the current model

The null deviance is

$$D(\text{null}) = 2 \{ \tilde{\ell}(\text{saturated}) - \tilde{\ell}(\text{null}) \}$$

where $\tilde{\ell}(\text{null})$ is the maximum of the log-likelihood under a model with a single parameter π

h) we want to test

$$\begin{cases} H_0: \beta_2 = \beta_3 = \beta_4 = 0 \\ H_1: \text{at least one is} \neq 0 \end{cases}$$

under H_0 , the model is $Y_i \sim \text{Ber}(\pi)$ $\text{logit}(\pi) = \beta_1$ "null model"

the maximum of the log-likelihood is $\tilde{\ell}(\text{null})$

under H_1 , I have the complete model

the maximum of the log-likelihood is $\hat{\ell}(\text{model})$

The LR test for testing the model is

$$W = 2(\hat{\ell}(\text{model}) - \tilde{\ell}(\text{null})) \sim \chi_{p-1}^2 = \chi_3^2$$

$$\begin{aligned} D(\text{null}) - D(\text{model}) &= 2 \{ \tilde{\ell}(\text{saturated}) - \tilde{\ell}(\text{null}) - \tilde{\ell}(\text{saturated}) + \hat{\ell}(\text{model}) \} \\ &= 2 \{ \hat{\ell}(\text{model}) - \tilde{\ell}(\text{null}) \} \end{aligned}$$

Hence the observed value of the test is

$$w^{\text{obs}} = 964.52 - 675.14 = 289.38$$

$$\text{Reject region: } R = (\chi_{3; 1-\alpha}^2; +\infty)$$

I reject H_0

